

MLOps Learning Journey

A structured, hands-on roadmap from Python fundamentals to a deployable MLOps project. Built from real Kaggle notebooks — every example was written and tested in a live environment.

About

I'm transitioning into MLOps and Data Science, documenting my learning publicly to build a practical portfolio.

This repo tracks my progress week by week — from NumPy basics to a fully deployed ML pipeline.

Roadmap

Week	Topic	Status
1-2	NumPy — data layer foundations	 Complete

Week	Topic	Status
3-4	Pandas — structured data & cleaning	✅ Complete
5	scikit-learn — preprocessing & first models	🔄 In Progress
6	Flask / FastAPI — wrap model in an API	🕒 Upcoming
7	Docker — containerise the API	🕒 Upcoming
8	CI/CD + Deployment — GitHub Actions + Render/Railway	🕒 Upcoming

Course Notes

Self-written beginner guides with tester and MLOps perspectives on every topic.

File	Description
numpy_mlops_course.md	Array creation, reshaping, boolean filtering, broadcasting,

File	Description
	save/load + mini project
pandas_mlops_course.md	DataFrames, null handling, groupby, merging, encoding + ML-ready pipeline project
categorical-encoding-notes.md	OrdinalEncoder vs OneHotEncoder, rare/unknown category handling, <code>min_frequency</code> vs <code>max_categories</code> , NumPy slicing + <code>plt.scatter</code> refresher

Repo Structure

```

MLOps beginner friendly learning
|
├── README.md
|
├── courses/
|   ├── numpy_mlops_course.md
|   └── pandas_mlops_course.md

```

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|   └─ categorical-encoding-notes.md
|
|─ notebooks/
|   └─ numpy_practice.ipynb
|   └─ pandas_practice.ipynb
|   └─ scikit-learn practice → see
Kaggle link below
|
└─ projects/
    └─ (deployable project coming in Week
8)
```

Tools & Environment

- **Kaggle Notebooks** — free GPU, pre-installed libraries
- **Python 3** — NumPy, Pandas, scikit-learn, Flask/FastAPI
- **Docker** — containerisation
- **GitHub Actions** — CI/CD automation

Kaggle Notebooks

- [scikit-learn — preprocessing, encoding & first models](#)
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Connect

- **Linkedin:**
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Updated weekly. Follow along if you're on a similar journey.